

Spatial transcriptomics (ST) data measure gene expression while preserving the spatial location of cells or spots within a tissue. In simple terms, it tells you which genes are expressed, and exactly where in the tissue they are expressed.

In this assignment, you will use 10x genomics visium dlpfc data.

Task 1: Read Visium Data - 4 marks

Here is a source where you can find the raw data:

https://drive.google.com/drive/folders/1Lz3_Bjct5pwf50Zrkt-uONNNdDFT182R?usp=sharing

```
visium_sample/  
├─ metadata.tsv  
├─ filtered_feature_bc_matrix.h5  
├─ 151671_visium_processed.h5ad  
└─ spatial/  
    ├─ tissue_positions_list.csv  
    ├─ scalefactors_json.json  
    └─ tissue_lowres_image.png  
        └─ tissue_hires_image.png
```

You will have to use **scanpy** to read data. You will read raw data and create an **AnnData object**.

The spots are given as barcodes and locations are given in `tissue_positions_list.csv`. The tissue image is also given. The `filtered_feature_bc_matrix.h5` contains gene expression data per spot. The annotations/ground truth for each spot is given in `metadata.tsv`. The column is called '**layer_guess_reordered**' (cluster type or spot type).

You can save the annotations or ground truth along with the AnnData object.

A processed AnnData Object for this data is already given in "151671_visium_processed.h5ad". You can test if you have read the annotations correctly or not by comparing your object with this stored object.

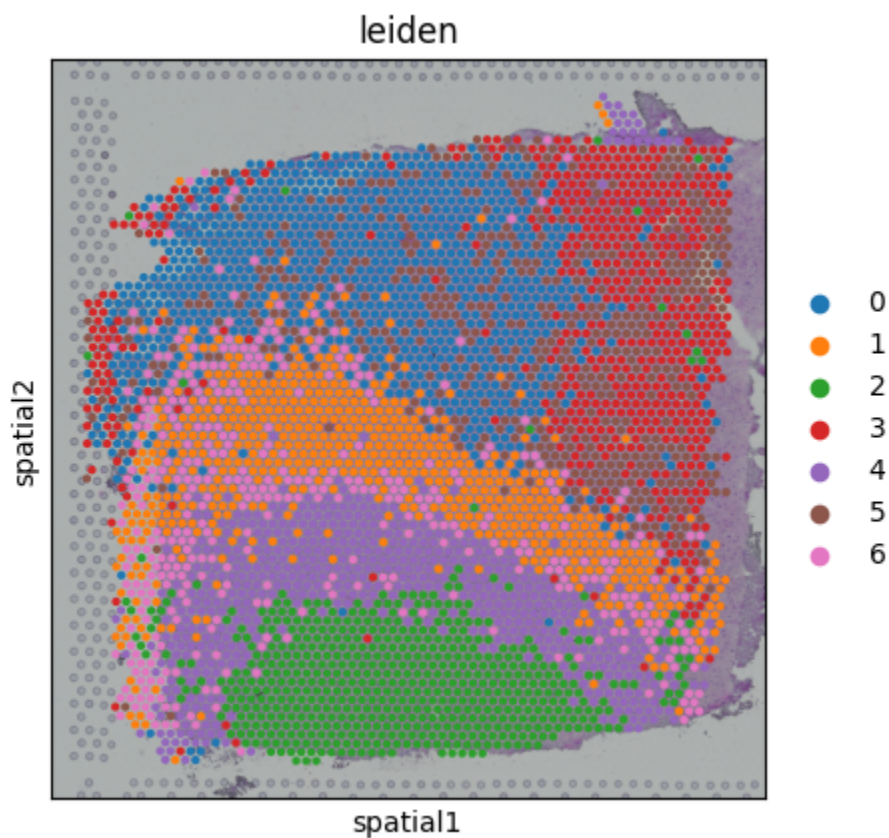
Task 2: Perform Clustering and visualize - 3 marks

Again, use scanpy and run the Louvain/Leiden algorithm and visualize the clustering using the hires tissue image. You have two options for this:

1. Perform PCA and then Louvain/Leiden -> visualize
2. Perform feature selection using seurat or any other method (wrapper filter) and then Louvain/Leiden -> visualize

For each of the cases report two metrics: ari and nmi. You may need to use scikit-learn and ground truth or annotations for this.

Sample leiden clustering visualization using scanpy:



Task 3: Fill up the following table - 3 marks

	After PCA		With Top 2000 features		With Top 5000 features	
dataset	ARI	NMI	ARI	NMI	ARI	NMI
151671						
151507						
151508						
151509						
151510						

You may find the other datasets here:

https://figshare.com/articles/dataset/10X_Visium/28200299

For each of them, produce clustering images on tissue.